

ML Graded Quiz M1

Question 1

When implementing gradient descent on ridge regression:

$$\theta^{[next\ step]} = \theta^{[previous\ step]} - \eta \nabla_{\theta} MSE(\theta^{[previous\ step]}) - \alpha \odot \theta^{[previous\ step]}$$

what does α represent?

- ☐ Vector containing the intensity of the penalty term for regularization
- ☐ Transition parameter from one step of the gradient descent to the next
- ☐ Gradient of MSE in the previous step
- ☒ Learning rate

Question 2

In the notebook from Lesson 4, we dealt with categorical variables that we had to transform for them to serve as inputs for the model. How did we treat education levels?

- ☐ We scaled the categorical value using min-max scaling.
- ☐ We created a numerical scale, where higher numbers corresponded to higher education levels.
- ☐ We created a numerical scale, where lower numbers corresponded to higher education levels.
- ☒ We built indicator variables for each education category with value equal to 1 if the instance belonged to that education category and 0 otherwise.

Question 3

You have designed 2 different prediction models, A and B. Model A has an in-sample R^2 of 65%, while the R^2 out of sample is 90%. Model B has an in-sample R^2 of 92%, while the out of sample R^2 is of 67%. According to these numbers, which model would you most likely use for prediction on a new dataset?

- ☐ Model B, since the in-sample R^2 is higher
- ☐ Model A, since the in-sample R^2 is lower
- ☐ Model B, since the out-of-sample R^2 is lower
- ☒ Model A, since the out-of-sample R^2 is higher

Question 4

Given the following 2x2 confusion matrix, what is the number of true negatives?

$$\begin{pmatrix} 1025 & 982 \\ 32069 & 30058 \end{pmatrix}$$

- ☐ 1,025
- ☐ 982
- ☐ 32,069
- ☒ 30,058

Question 5

In the penalty term associated with elastic net, what does λ represent?

- ☐ The higher λ , the higher the importance of parameter shrinkage in regularization
- ☐ The higher λ , the lower the importance of parameter selection in regularization
- ☐ The importance of regularization for the overall model
- ☒ Balance between the Ridge and LASSO-type penalty terms

Question 6

You are developing an ML classification model that aims to classify instances, given some features, into three categories (0, 1, and 2, in that order). You have obtained the following confusion matrix:

$$\begin{pmatrix} 1259 & 2 & 12 \\ 32 & 3058 & 155 \\ 59 & 97 & 2069 \end{pmatrix}$$

How many instances did the model classify into category 2 that actually belong to 1?

- ☐ 2
- ☐ 155
- ☐ 1,259
- ☒ 2,069

Question 7

In the following linear regression model:

$$\hat{y} = g(X) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

How many input features are there?

- ☐ 3
- ☐ 4
- ☐ There are 0 input features
- ☒ 2

Question 8

One scaling technique aims to produce de-correlated variables with unit variance. What is this technique called?

- ☐ Min-max scaling
- ☐ Normalization
- ☐ Whitening
- ☒ Standardization

Question 9

What is the k-fold cross-validation technique primarily used for?

- ☐ Validation
- ☐ Enhancing regularization
- ☐ Reducing model complexity
- ☒ Model testing

Question 10

You are developing an ML classification model that aims to classify instances, given some features, into three categories (0, 1, and 2, in that order). You have obtained the following confusion matrix:

$$\begin{pmatrix} 3987 & 2 & 12 \\ 32 & 3058 & 155 \\ 59 & 97 & 2069 \end{pmatrix}$$

How many instances did the model correctly classify into category 1?

- ☐ 2,069
- ☐ 97
- ☐ 3,058
- ☒ 59

Question 11

Which of the following expressions represents the R^2 measure?

☐ $R^2 = 1 - \sum_{i=1}^m (y^{(i)} - \bar{y})^2$

☐ $R^2 = 1 - \frac{\sum_{i=1}^m (y^{(i)} - x^{(i)}\hat{\theta})^2}{\sum_{i=1}^m (y^{(i)} - \bar{y})^2}$

☐ $R^2 = \frac{\sum_{i=1}^m (y^{(i)} - x^{(i)}\hat{\theta})^2}{\sum_{i=1}^m (y^{(i)} - \bar{y})^2}$

☒ $R^2 = 1 - \frac{\sum_{i=1}^m (y^{(i)} - \bar{y})^2}{\sum_{i=1}^m (y^{(i)} - x^{(i)}\hat{\theta})^2}$

Question 12

Which of the following confusion matrices represents a perfect classifier?

☐ $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

☐ $\begin{pmatrix} 2000 & 1000 \\ 1000 & 600000 \end{pmatrix}$

☐ $\begin{pmatrix} 0 & 982 \\ 32069 & 0 \end{pmatrix}$

☒ $\begin{pmatrix} 1025 & 0 \\ 0 & 30058 \end{pmatrix}$

Question 13

Which of the following is a penalty term to the cost function that ridge regression would use?

☐ $\alpha \sum_{i=1}^n |\theta_i|$

☐ $r\alpha \sum_{i=1}^n |\theta < em >_i| - \frac{1-r}{2}\alpha \sum_{<} /em >_i = 1^n \theta_i$

☐ $\alpha \sum_{i=1}^n \theta_i^2$

☒ $r\alpha \sum_{i=1}^n |\theta < em >_i| + \frac{1-r}{2}\alpha \sum_{<} /em >_i = 1^n \theta_i^2$

 Time left:
1:59:34

QUESTIONS

- | | | | |
|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Question 14

You are working on a binary classification model that yields the following confusion matrix:

$$\begin{pmatrix} 2648 & 0 \\ 689 & 1098 \end{pmatrix}$$

What is the F_1 score of the model?

- ☐ 100%
- ☐ 89.35%
- ☐ 76.11%
- ☒ 61.44%

Question 15

In the classic gradient descent equation:

$$\theta^{[next\ step]} = \theta^{[previous\ step]} - \eta \nabla_{\theta} MSE(\theta^{[previous\ step]})$$

What does η represent?

- ☐ Penalty term
- ☐ Updated estimate of θ_1
- ☐ Bias term
- ☒ Learning rate

Question 16

In a dual classification problem (i.e., two labels) of firm bankruptcy prediction, the number of true positives refers to what?

- ☐ Cases where the model predicted bankruptcy, and bankruptcy occurred
- ☐ Cases where the model predicted bankruptcy, but bankruptcy did not occur
- ☐ Cases where the model did not predict bankruptcy, and bankruptcy occurred
- ☒ Cases where the model did not predict bankruptcy, and bankruptcy did not occur

SUBMIT